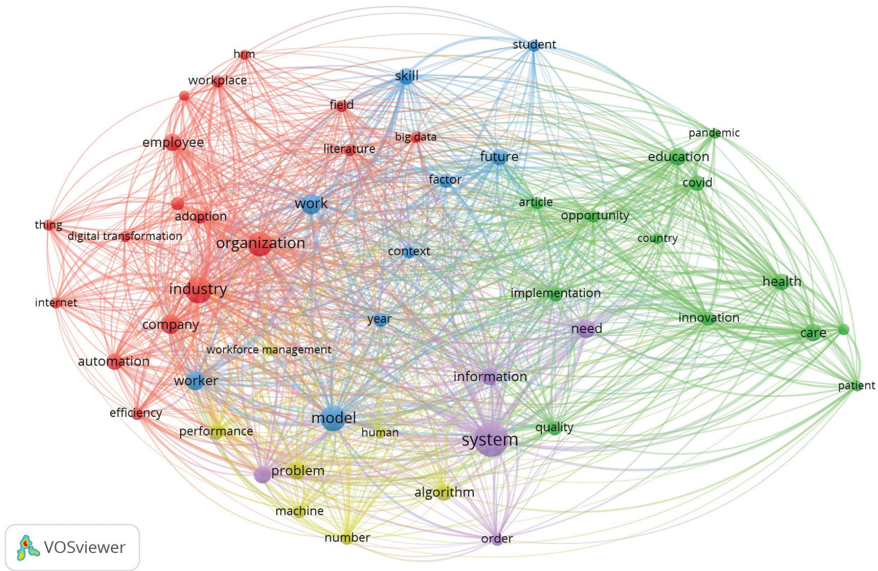

FIGURE 14.6 Co-occurrence author keywords network in the field of medicine and education.

***Source*: Author's own analysis from VOSviewer 1.6.18 (Nees Jan van Eck and Ludo Waltman).**

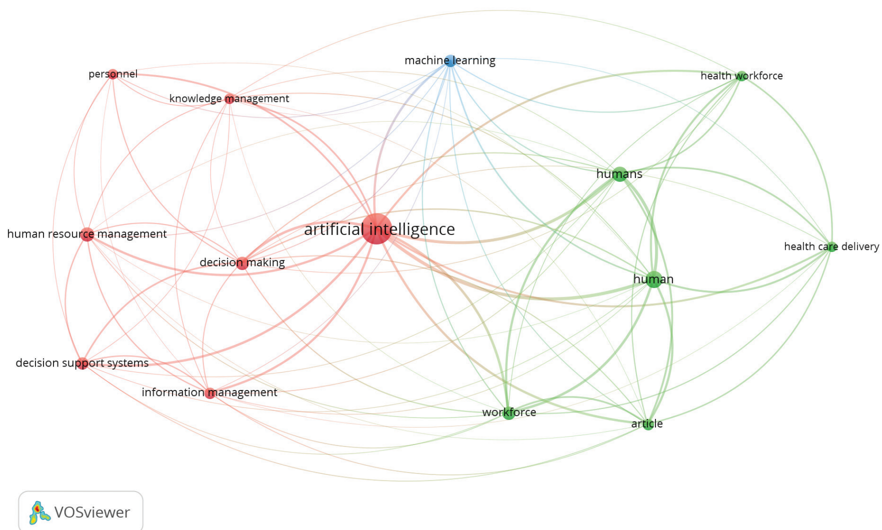

FIGURE 14.7 Co-occurrence author keywords network.

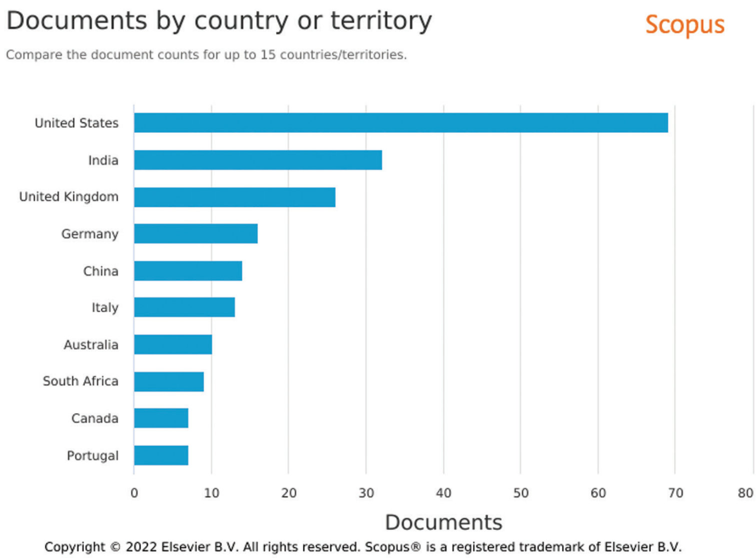

FIGURE 14.8 Documents by countries.

frequency of occurrence in the chapters is their connection weight. Three colors—red, blue, and green—distinguish terms in one cluster from those in other clusters, and the node's dimension reflects the total link strength. In the section that follows, the term "clusters" is examined in order to discuss the most pertinent research fads in the literature.

In the above visualization, the red cluster shows artificial intelligence, human resource management, decision-making, information management, etc. The green cluster shows healthcare delivery, human, workforce, health workforce, etc. as the most frequent items.

The blue-gray cluster includes machine learning in the visualization from the VOSviewer. There is an essential conclusion which could be empirically studied on the healthcare industry and the adoption of AI in their workforce, because many clusters reported healthcare and AI as connected nodes which deciphers there is a huge literature available, this also leaves us thinking about the gap which is evidently available in many public sectors, and the lack AI implementation especially in underdeveloped and developing countries.

14.5.3 Countries Network Analysis (VOS Clustering)

In countries including the United States, India, the United Kingdom, Germany, China, Italy, and Australia, where the topic was originally explored and is currently the subject of in-depth research, several academic scholars have made substantial

COUNTRY	DOCUMENTS
United States	69
India	32
United Kingdom	26
Germany	16
China	14
Italy	13
Australia	10
South Africa	9
Canada	7

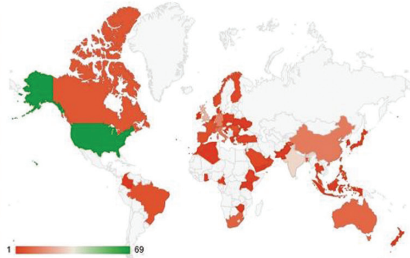

Figure 14.9 Visualization of countries producing research on AI and workforce management on a scale red to green.

FIGURE 14.9 Documents by countries.

contributions to the study of application of artificial intelligence in workforce management, especially in the healthcare field as shown in Figure 14.8.

The US ranks first, India ranks second, and the UK ranks third in the number of documents relating to research on artificial intelligence in workforce management, as shown in Figures 14.9 and 14.10.

Artificial intelligence has been the subject of most of the conceptual studies to date because developing and underdeveloped nations frequently imitate the behaviors of developed nations, enhancing the quality of interactions with researchers and having a significant positive impact on the overall body of technical knowledge on artificial intelligence.

According to this point of view, developed countries continue to have a significant influence on how researchers from developing countries view artificial intelligence and automation in general, how it is supported and funded by their governments, and how they are able to use in order to accomplish their goals of contributing to the workforce and industries at large.

14.5.4 Sponsoring Organizations Network Analysis (VOS Clustering)

There have also been several investigations into antecedents, mediating, and regulating processes of artificial intelligence in many areas. They discovered that artificial intelligence helps to manage the workforce better (Tailor, Pareek & Khang, 2022). The majority of these investigations were also carried out in developed and developing nations, which both expressed interest and published numerous publications in Scopus as Figure 14.11.

Research has come from health science, nursing universities, medical information, precision in symptoms, and medicine as shown in Figure 14.11.

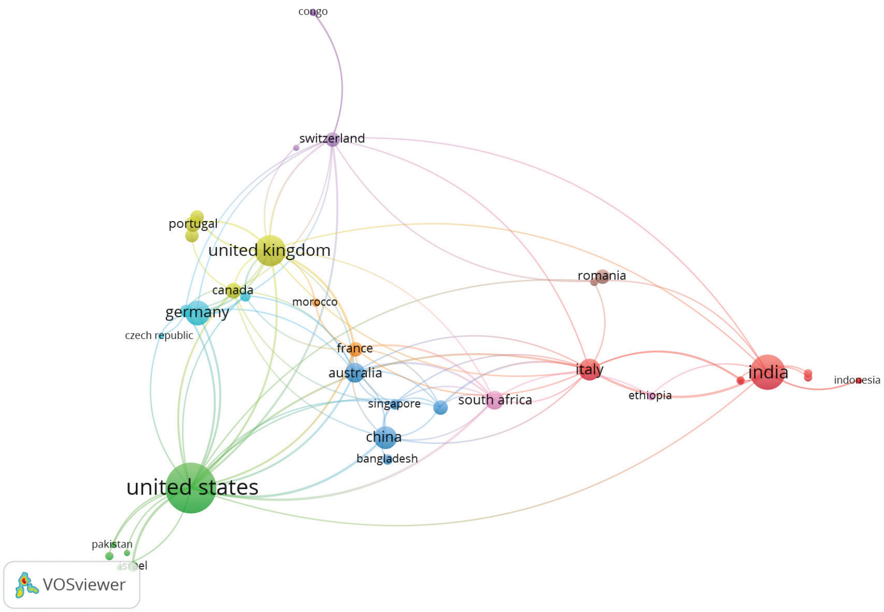

FIGURE 14.10 Documents by countries grouped by link strength.

***Source*: Author's own analysis from VOSviewer 1.6.18.**

14.6 LIMITATIONS

Only two keywords, "artificial intelligence" and "workforce management," were used, and even though they were relevant to the study's topic, the authors chose not to include any more. Had there been more keywords, the study would have been more comprehensive.

The study used information from the Scopus database, and other libraries with physical volumes that aren't published online could provide additional information for this study if they were accessible. Moreover, Web of Science and PubMed were not included in the scope of the study.

14.7 CONCLUSION

Numerous research has been conducted with AI from health science, nursing universities, medical information, precision in symptoms, and medicine. Artificial intelligence, rather than conventional completely model-based strategies, is roused by a "crude" discernible way of behaving instead of suppositions or total information.

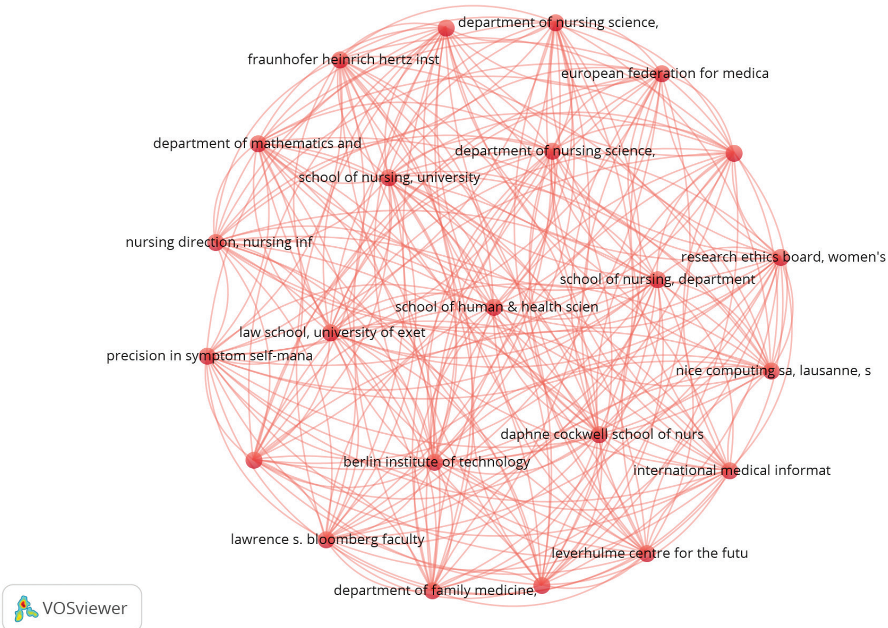

FIGURE 14.11 Visualization of organizations.

***Source*: Author's own analysis from VOSviewer 1.6.18.**

AI is completely process-situated and connects occasions to significant-level start-to-finish process models, as opposed to traditional information-driven methods. The gap between performance strategies like AI in workforce management can be applied. It looks for loopholes between naturally occurring or human-made mistakes (Shah et al., 2014).

In addition, difficult investigation are currently conceivable on the grounds of the advancement of interaction methods and how machine learning is applied in the area of healthcare in order to understand patient and nurse interaction, where they can understand the assistance needed to patient letting AI learn the ways to assist a patient, sooner or later AI will become the work-force itself, monitored by few guidelines and rules as well (Babasaheb et al., 2023a).

Artificial intelligence in human resource management systems (HRMS) processes for research is not a new subject. In fact, there have been a number of evaluations written about HRMS in general. These evaluations, nevertheless, do not emphasize AI (Khang et al., 2023c).

Since developing and underdeveloped countries frequently imitate the behaviors of developed countries, it has improved the quality of interactions with researchers and had a significant positive impact on the body of technical knowledge on artificial intelligence. This has led to artificial intelligence being the focus of the majority of conceptual studies to date (Khang et al., 2022d).

This viewpoint holds that developed nations continue to have a significant impact on how researchers from developing nations perceive automation and artificial intelligence in general, how it is supported and funded by their governments, or how they are able to work on it in order to achieve their goals of making a contribution to the workforce and industries at large (Misra et al., 2023). Regarding AI and its use in the fields of healthcare and education, there is a wealth of research and literature available.

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15 Leveraging Employee Data to Optimize Overall Performance

Using Workforce Analytics

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CONTENTS

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